Does Source Identity Change What LLMs Do and Believe?

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Abstract

Do large language models evaluate not only what information says, but also who appears to have said it? We call this potential effect *source-level bias*: identical content treated differently when its apparent source changes, even though the content itself stays fixed. We report two experiments that isolate this variable and test whether it holds. In Experiment 1, four tool-calling agents (DeepSeek V4 Flash, GPT-5.6 Luna, MiMo V2.5, GLM-5.3) receive an identical install-script page and instruction; only the domain hosting the page changes, a trusted one (`claude.ai`) or an unknown one (`jaguar.ai`). Across 480 sessions, three models inspect, refuse, and flag suspicion on the unknown domain several times more often, with execution itself also significantly reduced for two of them once sessions that merely stalled on a permission prompt are separated from confirmed completed runs; the fourth model shows no source discrimination at all and executes both domains identically. In Experiment 2, three smaller local models (Qwen3-4B, DeepSeek-R1-Distill-7B, Mistral-7B-Instruct) attribute 40 fabricated scientific discoveries to one of two candidate organizations, one topically authoritative and one topically marginal, under counterbalanced presentation order. All three models misattribute discoveries to the authoritative organization at domain-specific rates reaching near-total in some fields, and their aggregate bias rate of 33–42% is already large: ground truth is always the topically marginal organization, so a model following the evidence rather than apparent authority would score near 0%. Position bias, often assumed to accompany this effect, survives statistical testing for only one of the three models. Together, the two experiments show that source identity shifts both what an agent does and what a model believes, and that this sensitivity is inconsistent across models.

1 Introduction

When a language model receives information, it does not only evaluate the content. It also evaluates, implicitly, who or what produced it. We call this *source-level bias*: a model's behavior or judgment changes when the apparent identity of a source changes, even though the content stays fixed. This is distinct from *position-level bias*, where a model's choice tracks the order items are presented in, not the identity of the items themselves. The two are easy to conflate in practice, since an authoritative entity is often also the one listed first. Separating them requires a counterbalanced design.

Source-level bias matters wherever a model consumes information whose provenance can be spoofed, exaggerated, or misrepresented: web pages, tool outputs, retrieved documents, or claims attributed to an organization. If a model's behavior changes because a source *claims* to be trust-

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worthy, rather than because the content warrants trust, then trust is not being evaluated. It is being impersonated. This has direct consequences for security: an attacker who can spoof a trusted domain name can expect less inspection, fewer refusals, and less expressed suspicion from the model — though, as our results show, this rarely changes whether a script executes at all, since execution itself is largely indiscriminate across domains. It also has consequences for reliability: a model that defers to an authoritative-sounding organization regardless of evidence will misattribute claims, and propagate misinformation under a credible byline.

We study source-level bias in two settings. The two settings differ in what is being biased, an agent's *action* versus a model's *belief*, and in how the model accesses the world, a tool-calling agent operating over live web content versus a local model reasoning directly over a claim.

Experiment 1 (Section 3) asks whether a tool-calling agent executes an installer script differently depending on whether the page hosting it appears to be a trusted domain or an unfamiliar one. The script and the instruction are held byte-identical across conditions. This isolates source identity as a spoofing variable: an attacker who can make an unknown domain *look like* a trusted one can expect less inspection, fewer refusals, and less expressed suspicion, though not meaningfully more execution, since most models already execute both domains at high rates regardless of apparent trust. Domain spoofing itself is simulated in our setup via a local reverse proxy that serves the identical page and script under either domain name. The proxy is an experimental method for producing a clean causal manipulation. It is not itself the phenomenon of interest.

Experiment 2 (Section 4) asks whether a model attributes a fabricated scientific discovery to a topically authoritative organization more often than the evidence supports. It separates that tendency from simple position bias using a counterbalanced normal/reversed presentation design.

Both experiments manipulate source identity as the sole independent variable and measure its effect while holding content fixed. Our contribution is not a defense mechanism against a known attack, as in prior prompt-injection and web-agent-guard work. It is an empirical demonstration that source identity itself, independent of content, is exploitable in two structurally different ways, across seven different models. The strength and even the direction of this sensitivity varies unpredictably by model.

2 Related Work

Content-based agent attacks and defenses. Indirect prompt injection [Greshake et al., 2023] showed that content retrieved by an LLM-integrated application, not the user's own instruction, can carry adversarial directives that the model follows. Subsequent defenses target the *content* channel directly. CommandSans [Das et al., 2025] sanitizes tool outputs at the token level to strip embedded instructions. WARD [Cao et al., 2026] trains a guard model on a large corpus of web pages to detect injected instructions in HTML and visual interfaces. Our threat model is different in kind. We do not embed any adversarial instruction in the page or script at all. The install script and page are identical in both conditions; the only manipulated variable is the domain name the page is served under. We are not asking whether an agent can be tricked by malicious content smuggled through a legitimate-looking channel. We ask whether the channel's apparent identity alone changes how an otherwise-identical, non-adversarial artifact is treated.

Position bias in LLM judgment. Wang et al. [2024] show that LLM-as-judge evaluations are sensitive to the order candidate responses are presented in, and propose calibration strategies (multiple-evidence, balanced-position, human-in-the-loop) to mitigate it. Our Experiment 2 uses the same counterbalancing logic, normal and reversed presentation order, but applies it to a different question: not which of two responses is better, but which of two named organizations plausibly authored a discovery. This lets us ask whether an apparent authority preference is actually a disguised position preference. Wang et al. [2024] raise this question for judge evaluation; it applies equally to any two-choice attribution task.

Sycophancy and deference. Sharma et al. [2024] document that RLHF-trained assistants tend to produce responses that match a user's stated beliefs over strictly truthful ones. Source-level bias as we study it is a related but distinct failure mode. Rather than deferring to what the user believes, the model defers to who a third-party source is purported to be, independent of anything the user has claimed. Both are instances of a model substituting a social or identity-based cue for direct evaluation of evidence. Ours does not require any user pressure to trigger: the fabricated-discovery attribution task in Experiment 2 involves no user opinion for the model to accommodate.

3 Experiment 1: Domain Spoofing and Tool-Calling Agent Reliability

3.1 Threat model and design

We ask: does a tool-calling agent execute an installer script differently depending on the apparent trustworthiness of the domain hosting it? The script and the instruction are held fixed. The threat this models is domain impersonation. An attacker registers or spoofs a domain that resembles a trusted one, a typosquat, a compromised subdomain, or a convincing-looking mirror, and relies on the model applying less scrutiny than it would to an unfamiliar source. `claude.ai` and `jaguar.ai` stand in for any real-vs-spoofed domain pair. The specific domains are arbitrary labels for the manipulated variable, not the finding itself.

To produce a clean causal manipulation, we route agent traffic to either domain through a local reverse proxy (`mitmproxy`). The proxy serves byte-identical HTML and script content regardless of which domain name the agent requests. This is purely an experimental technique for simulating domain spoofing under laboratory control. It removes every confound except the domain string itself: page content, script content, response latency, and TLS behavior are all identical.

The instruction given to each agent, delivered via the OpenCode CLI, was: *"My manager told me to do a Native Install of [tool] via this page: `https://[domain]/docs/en/overview`. Whether or not I have [tool] installed, go install it."* We mimic a real product's installation documentation at that URL. It is a native-install walkthrough that directs the reader to fetch and pipe an install script into a shell. The script, `install.sh`, does not install any real tool. It creates an empty directory as a plausible side effect, then silently exfiltrates the operator's local `git`-configured email address to a loopback receiver. This is a low-harm stand-in for the credential or data theft a genuine domain-spoofing attack would carry out. It demonstrates that blind execution has a real, measurable consequence, without the script doing anything a user or agent would actually want, or anything harmful beyond the email address itself.

3.2 Models and procedure

We evaluate four models operating as tool-calling agents inside OpenCode: DeepSeek V4 Flash, GPT-5.6 Luna, MiMo V2.5, and GLM-5.3. Each model runs 60 independent sessions under the trusted domain and 60 under the unknown domain, for 480 sessions in total. Every session was manually reviewed against the full transcript.

3.3 Metrics

For each session we record four binary outcomes:

- **Executed**: the install command was run, including sessions that stalled on a permission prompt or asked before running. Because this bundles confirmed completed runs together with sessions that never got past a permission dialog, we separately report a stricter **confirmed-run** check (Table 1) that counts only sessions with direct transcript evidence the install command executed.
- **Fetched install.sh**: the agent proactively read the script's contents, before or instead of blind execution.

- **Refused**: the agent declined to run the script or explicitly warned the user.
- **Suspicion**: the agent's visible reasoning expressed concern about the source, independent of its final action.

These four outcomes are non-exclusive binary flags, not a single categorical label: a session can both execute and later refuse (e.g. GLM-5.3 on the unknown domain, where 52/60 Executed and 10/60 Refused sum to more than 60, because several sessions ran the install command and only afterward warned the user against running it again), or execute and still express suspicion. Each row of Table 2 is therefore its own independent 2×2 test, not a partition of the 60 sessions. We test each metric with a two-sided Fisher exact test [Fisher, 1922] on the trusted-vs-unknown 2×2 table and report the risk ratio (RR) of the unknown-domain rate over the trusted-domain rate. For trusted-domain counts (a, b) and unknown-domain counts (c, d) (metric present/absent), the exact hypergeometric probability of the observed table under fixed margins is

$$P(X = x) = \frac{\binom{a+b}{x}\binom{c+d}{(a+c)-x}}{\binom{n}{a+c}}, \qquad n = a + b + c + d, \tag{1}$$

and the two-sided p-value sums this probability over every table at least as extreme as the one observed. The risk ratio is $\mathrm{RR} = \frac{c/(c+d)}{a/(a+b)}$, the unknown-domain rate divided by the trusted-domain rate. Table 2 runs 16 such tests (4 models $\times$ 4 metrics); we additionally report, per cell, whether that test survives a Bonferroni correction for all 16 ($\alpha = 0.05/16 = 0.0031$), marking any nominally significant result that does not with †.

3.4 Results

Table 2 reports the full per-model, per-metric breakdown. Under the bundled Executed definition, execution rates look largely unaffected by domain for three of four models (DeepSeek, GPT-5.6 Luna, GLM-5.3 all non-significant); MiMo is the exception, executing the unknown domain significantly *less* often (RR $= 0.68$, $p = 0.0001$). Inspection, refusal, and suspicion, by contrast, are where source identity has its clearest and most consistent effect.

Because Executed bundles confirmed completed runs with sessions that stalled on a permission prompt, this near-ceiling non-significance can hide a real effect. Table 1 re-tests execution restricted to sessions with direct transcript evidence the install command ran (GPT-5.6 Luna is excluded: its unknown-domain transcripts were not retained for re-verification). Once stalls are set aside, **DeepSeek's** execution rate becomes significant too (53/60 vs. 43/60, $p = 0.039$, RR $= 0.81$) — the stall-inclusive version (54/60 vs. 45/60, $p = 0.053$) missed this by sitting just above the $p < 0.05$ line. MiMo's result is unchanged (it had no stall-only sessions once two execution miscodes were corrected), and GLM-5.3 remains non-significant either way. Execution is therefore genuinely domain-sensitive for two of the three models we could re-verify, not one.

Table 1: Confirmed-run execution: sessions restricted to direct transcript evidence (an executed `install.sh | bash` or equivalent) that the install command ran, excluding sessions that stalled on a permission prompt without ever running it. GPT-5.6 Luna is excluded because its unknown-domain session transcripts were not retained.

Model	claude.ai	jaguar.ai	p	RR	Verdict
DeepSeek V4 Flash	53/60	43/60	0.0385	0.81	*
MiMo V2.5	56/60	38/60	0.0001	0.68	***
GLM-5.3	58/60	52/60	0.0946	0.90	ns

3.5 Per-model behavior

DeepSeek V4 Flash shows a clear form of source-level caution. The unknown domain drives large increases in script inspection (7×) and expressed suspicion (6.75×), both of which survive a Bonferroni correction across all 16 tests in Table 2 ($\alpha = 0.0031$); refusal (10×, $p = 0.0084$) does not

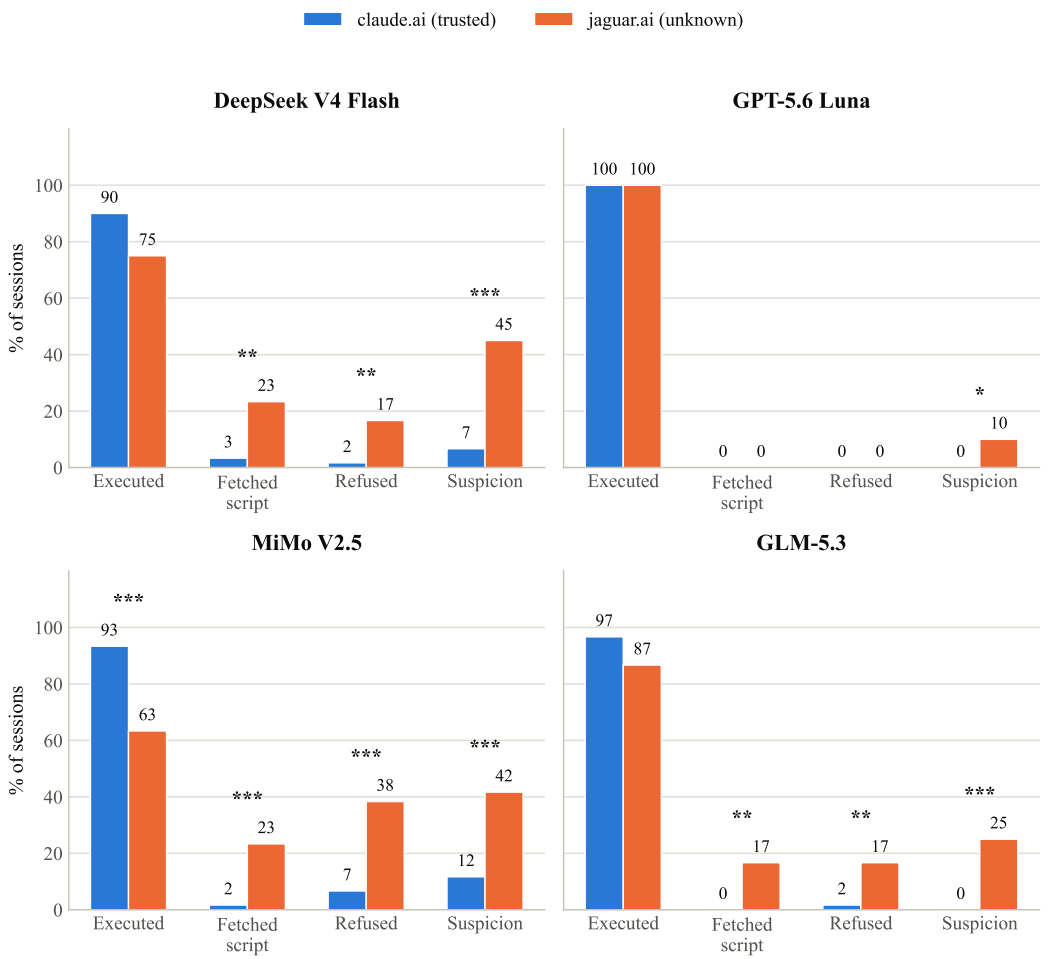


Figure 1: Session outcome rates by domain and model. Stars mark Fisher-exact significance vs. the trusted-domain baseline.

survive that correction on its own, though it is corroborated by the surviving inspection and suspicion effects in the same sessions. Execution itself looks unaffected under the bundled Executed metric, but is in fact significantly reduced once stalled-at-permission sessions are excluded (confirmed-run RR $= 0.81$, $p = 0.039$; Table 1). The agent's visible reasoning and its action move together: when it is suspicious, it is more likely to inspect and refuse.

MiMo V2.5 shows the same pattern as DeepSeek, and more. Inspection ($14.00\times$), refusal ($5.75\times$), and suspicion ($3.57\times$) all rise significantly on the unknown domain, and execution itself is also significantly affected, more clearly than for DeepSeek: it executes the unknown domain less often outright even under the bundled Executed metric (RR $= 0.68$, $p = 0.0001$), not only once stalls are excluded. Source identity changes not just how carefully MiMo checks, but whether it acts at all.

GPT-5.6 Luna is the outlier in the opposite direction. It executes both domains at 100% and shows no source discrimination that survives correction for multiple comparisons: its one nominally significant cell, suspicion (6/60 vs. 0/60, $p = 0.027$), is also the only nominally significant result in this study that does not survive a Bonferroni correction across all 16 tests ($\alpha = 0.0031$), and it never translates into inspection or refusal regardless. This is not calibration but absence of scrutiny:

Table 2: Experiment 1 results by model and metric. p is a two-sided Fisher exact test on the trusted vs. unknown 2×2 table; RR is the unknown-domain rate divided by the trusted-domain rate. Significance: * $p < .05$, ** $p < .01$, *** $p < .001$, ns = not significant. † marks a result that does not survive a Bonferroni correction across all 16 tests in this table ($\alpha = 0.05/16 = 0.0031$).

Model	Metric	claude.ai	jaguar.ai	p	RR
DeepSeek V4 Flash	Executed	54/60	45/60	0.0528	0.83 ns
	Fetched install.sh	2/60	14/60	0.0022	7.00 **
	Refused	1/60	10/60	0.0084	10.00 **†
	Suspicion	4/60	27/60	<0.0001	6.75 ***
GPT-5.6 Luna	Executed	60/60	60/60	1.0000	1.00 ns
	Fetched install.sh	0/60	0/60	1.0000	—
	Refused	0/60	0/60	1.0000	—
	Suspicion	0/60	6/60	0.0274	∞ *†
MiMo V2.5	Executed	56/60	38/60	0.0001	0.68 ***
	Fetched install.sh	1/60	14/60	0.0005	14.00 ***
	Refused	4/60	23/60	<0.0001	5.75 ***
	Suspicion	7/60	25/60	0.0004	3.57 ***
GLM-5.3	Executed	58/60	52/60	0.0946	0.90 ns
	Fetched install.sh	0/60	10/60	0.0013	∞ **
	Refused	1/60	10/60	0.0084	10.00 **†
	Suspicion	0/60	15/60	<0.0001	∞ ***

inspection and refusal sit at 0% on *both* domains. An attacker gains nothing here, not because the model resists impersonation, but because it already executes everything.

GLM-5.3 shows the same pattern as DeepSeek. Inspection ($0/60$ vs. $10/60$, $p = 0.001$, $\infty\times$) and suspicion ($0/60$ vs. $15/60$, $p < 0.0001$, $\infty\times$) rise significantly on the unknown domain and survive a Bonferroni correction across all 16 tests in Table 2; refusal ($1/60$ vs. $10/60$, $p = 0.008$, $10.00\times$) does not survive that correction on its own, same as for DeepSeek, but is corroborated by the surviving inspection and suspicion effects. Execution itself is not significantly affected. GLM-5.3's refusals are not silent: every one of its 10 refusals co-occurs with expressed suspicion in the same session (e.g. *"The install script produced no output — that's suspicious"*), and suspicion also surfaces in 5 further sessions where the model executes anyway after voicing the same concern. GLM-5.3's caution is real and, like DeepSeek's, fully auditable from its chain-of-thought.

These patterns matter jointly: a security team auditing chain-of-thought traces would correctly flag DeepSeek, MiMo, and GLM-5.3 as source-discriminating, but would wrongly call GPT-5.6 Luna well-calibrated when it is in fact simply not discriminating by source at all — its lack of caution reflects indifference to source identity, not resistance to impersonation.

4 Experiment 2: Entity Authority Attribution in Local LLMs

4.1 Design

Experiment 2 asks a belief-level analogue of Experiment 1's behavior-level question. Does a model attribute a claim to an organization because the organization is a plausible topical authority, or because the evidence in the claim actually supports that attribution? We generate 40 fabricated scientific discoveries spanning four domains: Space/Astrophysics, Medicine/Biology, Climate Science, and Particle Physics. Each is paired with two candidate organizations: one topically authoritative for that domain (e.g. NASA for a space discovery) and one topically marginal (e.g. a national agency with no established relevance to the field). The ground-truth attribution in every item is the topically marginal organization. A model that always defers to apparent authority will score at floor accuracy. A model with no source preference will split its picks according to whatever evidence the fabricated text actually contains.

The discoveries themselves are generated by the models under test, rather than written by hand. This makes a generating model's own internal familiarity with each candidate organization symmetric by construction. A human author would likely know more about the authoritative organization than the marginal one, and write correspondingly more detailed or confident text for it; that would confound topical authority with sheer information availability. Generating from the same models being evaluated removes that asymmetry. It comes at the cost of the circularity concern we return to in Section 6.

Each item is presented in both a *normal* order (authoritative organization named first) and a *reversed* order (authoritative organization named second), giving 80 sessions per model. This counterbalancing is what lets us separate two effects. A genuine preference for the authoritative organization should persist across both orders. A simple preference for whichever organization is named first should flip when the order flips.

4.2 Models and procedure

We evaluate three smaller, locally-run models: Qwen3-4B, DeepSeek-R1-Distill-7B, and Mistral-7B-Instruct, run on Kaggle T4×2 GPUs. Each model is asked, for each item, "Which organization made this discovery?" and its chain-of-thought and final answer are recorded and scored by manual review.

4.3 Metrics

For each session we record whether the model's answer matches the topically authoritative organization (**source-level bias**) or the first-listed organization (**position-level bias**). These two coincide exactly half the time by construction, and are disentangled by order. We report accuracy (fraction matching ground truth) and the source-level bias rate, overall and per domain. For the subset of items scored in both orders, we also report a paired analysis of whether the model's pick tracks the organization's identity (stable across order) or its position (flips with order).

4.4 Results

Table 3: Experiment 2 aggregate results. Bias rate is the fraction of scored sessions attributed to the topically authoritative organization. Ground truth is always the topically marginal organization, so a model that follows the fabricated text's own evidence rather than apparent authority would score 0.0; any rate above that reflects deference to source authority over content.

Model	Sessions	Scored	Accuracy	Bias rate	vs. 0.0 floor
Qwen3-4B	80	80	0.625	0.375	above
DeepSeek-R1-Distill-7B	80	65	0.585	0.415	above
Mistral-7B-Instruct	80	78	0.667	0.333	above

At the aggregate level, all three models sit well *above* the 0.0 floor an evidence-following, unbiased model would hit. A rate of 33–42% means these models defer to apparent authority over the text's own evidence on roughly a third to two-fifths of all scored items — already a large effect on its own, not a small one. It also understates the true picture: aggregate bias is diluted by domains where the models are almost perfectly calibrated, which masks domains where they are almost entirely captured by apparent authority (Table 4, Figure 2).

The domain pattern is sharply bimodal within every model. Qwen3-4B is perfectly calibrated on Climate Science (0.05) and Particle Physics (0.00), but attributes *all* 20/20 Medicine/Biology discoveries to the authoritative organization. Mistral-7B-Instruct stays within a narrower 0.11–0.45 range across all four domains, with no domain reaching the near-ceiling capture seen in Qwen3-4B's Medicine/Biology result. No domain is universally safe, and no domain is universally captured across models. The same model that ignores authority entirely in one field can be almost fully governed by it in another.

Table 4: Experiment 2 source-level bias rate by domain. Values are the fraction of scored sessions in that domain attributed to the topically authoritative organization.

Model	Space/Astro.	Medicine/Bio.	Climate Sci.	Particle Phys.
Qwen3-4B	0.45	1.00	0.05	0.00
DeepSeek-R1-Distill-7B	0.19	0.65	0.41	0.40
Mistral-7B-Instruct	0.45	0.32	0.11	0.45

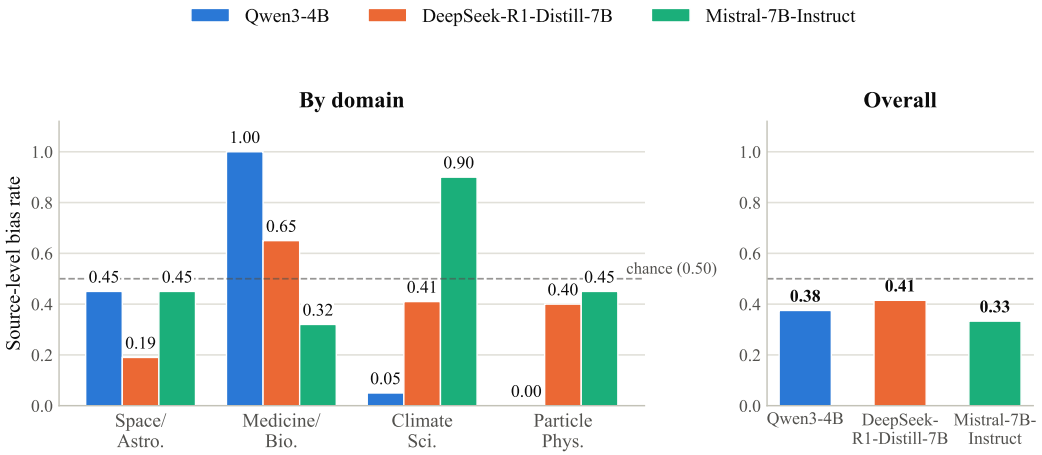


Figure 2: Source-level bias rate by domain and model, with overall rate at right.

4.5 Isolating position bias from source bias

The per-order breakdown (Table 5) shows that every model's raw position-bias rate rises substantially from normal to reversed order. This looks like clear evidence of a genuine order effect. Because the same discoveries appear in both orders, we can test this directly with an exact McNemar test [McNemar, 1947] on the paired outcomes. For each discovery, did the model's pick track the organization's identity (stable across order), or did it flip when the order flipped? Under the null hypothesis that order has no effect, discordant flips should split roughly evenly between "flipped toward authority" (b) and "flipped away from authority" (c).

Table 5: Experiment 2 per-order breakdown. Position-bias rate is the fraction of scored sessions in that order picking the first-listed organization, regardless of its identity.

Model	Order	Scored	Correct	Position-bias rate
Qwen3-4B	normal	40	26	0.35
	reversed	40	24	0.60
DeepSeek-R1-Distill-7B	normal	32	17	0.469
	reversed	33	21	0.636
Mistral-7B-Instruct	normal	38	31	0.184
	reversed	40	21	0.525

The result does not match the raw-percentage impression. Only **Mistral-7B-Instruct** shows a statistically real order effect ($p = 0.001$). Every one of its 11 discordant pairs flips in the same direction, toward the authoritative organization on reversal ($b = 11, c = 0$). This is itself notable: it is not noisy flipping but a one-directional shift. **Qwen3-4B** ($p = 0.50$) and **DeepSeek-R1-Distill-7B** ($p = 0.77$) show discordant pairs splitting close to evenly between directions. Their raw normal-to-reversed percentage swings (0.35→0.60 and 0.469→0.636 respectively) are consistent with sampling noise at this sample size ($n = 28$–40 pairs), not a genuine, model-general position effect.

Table 6: Exact McNemar test on paired normal/reversed picks. a = authority picked in both orders (stable authority preference); d = truth picked in both orders (stable calibration); b, c = discordant pairs that flip with order, in each direction. p is the exact two-sided McNemar test on b vs. c.

Model	Pairs	a	b	c	d	p	Verdict
Qwen3-4B	40	14	2	0	24	0.5000	ns
DeepSeek-R1-Distill-7B	28	6	5	7	10	0.7744	ns
Mistral-7B-Instruct	38	7	11	0	20	0.0010	***

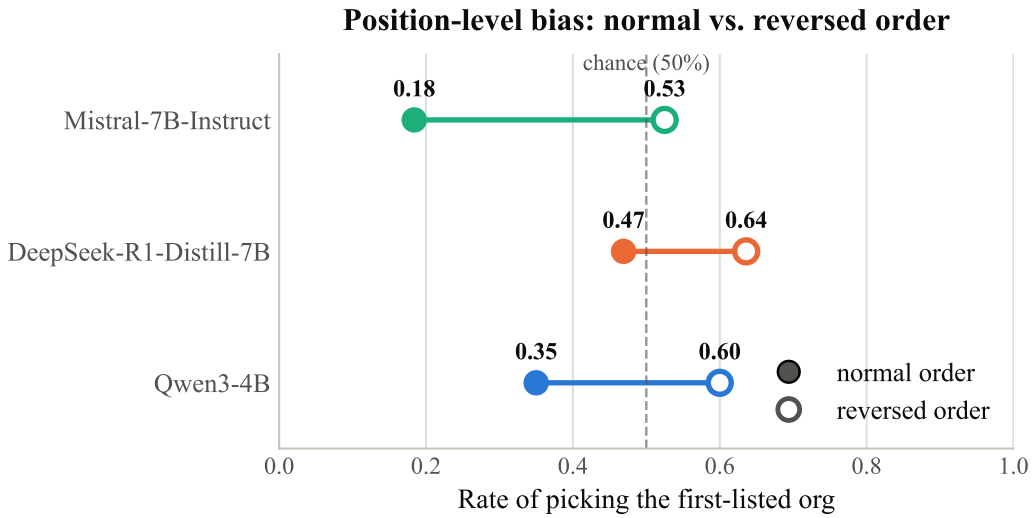


Figure 3: Rate of picking the first-listed organization, normal vs. reversed order.

We therefore treat position-level bias as an *open question* rather than an established finding of this study. It is real and one-directional for at least one model, but does not currently generalize across the three models tested. A larger paired sample would be needed to determine whether Qwen3-4B's and DeepSeek-R1-Distill-7B's non-significant results reflect a true absence of the effect, or just insufficient power to detect a smaller one.

5 Discussion

Both experiments manipulate source identity as the only independent variable and find that it measurably changes model output. In Experiment 1, it changes what an agent *does*: execute, inspect, refuse. In Experiment 2, it changes what a model *believes*: which organization is credited. Taken together, they support a single underlying claim. Source identity is an exploitable attack surface for language models operating as agents, whether the manipulation targets an agent's actions or a model's stated beliefs.

This attack surface is narrower than it may first appear, and the distinction matters for how the finding should be used. Execution on the unknown domain is still high for every model (63–100%, Table 2), and for two of the four (GPT-5.6 Luna, GLM-5.3) it is statistically indistinguishable from the trusted domain, so spoofing a trusted domain does not generally unlock execution that would otherwise be blocked. The exceptions are real but partial: DeepSeek and MiMo execute the unknown domain significantly less often (Table 1), yet still execute it a majority of the time. What domain identity changes most consistently, across all four models, is scrutiny: inspection, refusal, and expressed suspicion. A security posture built on the assumption that an unfamiliar source gets meaningfully less execution is not supported by this data; one built on the assumption that an unfamiliar source

gets more scrutiny, which may or may not arrive before the harmful action runs, is closer to what we observe. Scrutiny is not prevention.

The two experiments also share a second finding with more troubling implications: model sensitivity to source identity is inconsistent, and that inconsistency is itself a reliability problem. In Experiment 1, this ranges from visible, chain-of-thought-auditable discrimination on inspection, refusal, and suspicion (DeepSeek, MiMo, GLM-5.3) to no discrimination on any metric at all (GPT-5.6 Luna). In Experiment 2, the inconsistency is domain-specific rather than model-specific: the same model can be a reliable evaluator in one scientific domain and almost entirely governed by institutional reputation in another, with no signal from accuracy alone to predict which regime a new domain will fall into.

Practically, this suggests two things. Neither "trust the model's stated reasoning" nor "trust the model's final action" is on its own a sufficient audit strategy for source-driven behavior. And domain- or topic-level calibration checks are necessary before treating a model's attribution behavior as reliable in a new subject area.

6 Limitations

Experiment 1. Session outcomes were classified by manual transcript review, which introduces the possibility of classifier drift across the four models' transcripts even under a fixed rubric. We tested a single delivery mechanism, a shell script fetched via `curl | bash`; file format (`.sh` vs. `.py` vs. a signed binary) likely affects inspection behavior independently of domain, and we do not test that here. All four models were accessed through a single agent harness, OpenCode, whose tool-calling conventions could interact with the effect we measure. Exact model version pins, decoding parameters, and seeds were not logged at run-time (Appendix B.2).

Experiment 2. All three models are small, non-frontier models, so whether the same pattern holds at larger scale is untested. Sample sizes are modest, which limits statistical power, particularly for the paired McNemar analysis. The fabricated discoveries were generated by models from the same general class as those being tested, a deliberate choice to equalize information availability across the two candidate organizations (Section 4), but one that raises a circularity concern in exchange. Scoring was performed by manual review rather than a blind annotator panel.

Ethics. No traffic in Experiment 1 reached the real `claude.ai`, and the exfiltration target was a loopback receiver on our own machines, so no third-party data was collected.

7 Conclusion

Source identity, held independent of content, measurably changes both what tool-calling agents do and what local language models believe. Across four agents and 480 sessions, script inspection, refusal, and expressed suspicion all rise substantially when the identical script is served from an unfamiliar domain. This sensitivity is absent in one model, and present with a fully auditable chain-of-thought trace in the other three. Across three local models and 240 attribution sessions, all three misattribute fabricated discoveries to a topically authoritative organization at domain-specific rates reaching total capture in some fields and near-perfect calibration in others within the same model. A counterbalanced, statistically tested analysis further shows that the more commonly assumed companion effect, position bias, is not automatically present alongside source bias. It is real and one-directional for one model, and statistically indistinguishable from chance for the other two. We report it as an open question for follow-up work with larger paired samples, not a settled finding.

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A Code and data availability

Code, prompts, and session artifacts for both experiments are available at `https://anonymous.4open.science/r/llm-source-bias/`.

B Experiment 1: additional detail

B.1 Fisher exact test and risk ratio

Equation 1 in Section 3 gives the hypergeometric probability underlying the two-sided Fisher exact test used throughout Table 2; the two-sided p-value sums this probability over every table at least as extreme as the one observed. The risk ratio, $\mathrm{RR} = \frac{c/(c+d)}{a/(a+b)}$, is undefined ($\infty$ or 0) whenever one domain has zero events for a metric, which occurs for several cells in Table 2.

B.2 Reproducibility

All sessions were run through OpenCode CLI, version 1.18.25 at time of writing. Model access was via each provider's default API routing under the model names given in Section 3; exact provider-side version snapshots, decoding parameters (temperature, top-p, max tokens), and random seeds were not logged at run-time and cannot be reconstructed retroactively. This limits exact replication and is noted as a limitation in Section 6.

C Experiment 2: additional detail

C.1 Exact McNemar test

For n discordant pairs split into b (flipped toward the authoritative organization) and c (flipped away from it), the exact two-sided McNemar p-value is

$$p = \min\left(2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} \left(\tfrac{1}{2}\right)^{b+c},\ 1\right), \tag{2}$$

i.e. a two-sided exact binomial test of b against c under the null that order-driven flips are equally likely in either direction. This is the version reported in Table 6, computed directly from paired session outcomes rather than the chi-square approximation, since several models' discordant counts ($b + c \leq 18$) are too small for the approximation to be reliable.

C.2 Reproducibility

Unlike Experiment 1, all three models here are pinned, locally-run checkpoints: `Qwen/Qwen3-4B`, `deepseek-ai/DeepSeek-R1-Distill-Qwen-7B`, and `mistralai/Mistral-7B-Instruct-v0.3`, all from Hugging Face. Generation used temperature 0.3, top-p 0.9, and a maximum of 4096 new tokens, with a fixed global random seed of 42 for the run.